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RADIX SORT

PROGRAM CODING:

```
#include<stdio.h>

#include<conio.h>

radix_sort(int array[], int n);

void main()

{

int array[100],n,i;

clrscr();

printf("Enter the number of elements to be sorted:

");

scanf("%d",&n);

printf("\n Enter the elements to be sorted: \n");

for(i =0;i<n;i++ )

{

printf("\tArray[%d] = ",i);

scanf("%d",&array[i]);

}

printf("\n Array Before Radix Sort:"); //Array Before Radix Sort

for(i =0;i<n;i++)

{

printf("%8d", array[i]);

}

printf("\n");

radix_sort(array,n);

printf("\n Array After Radix Sort: "); //Array After Radix Sort
```

```
for(i =0;i<n;i++)
{
printf("%d", array[i]);
}
printf("\n");
getch();
}
radix_sort(int arr[], int n)
{
int bucket[10][5],buck[10],b[10];
int i,j,k,l,num,div,large,passes;
div=1;
num=0;
large=arr[0];
for(i=0; i<n; i++)
{
if(arr[i] > large)
{
large = arr[i];
}
while(large > 0)
{
num++;
large = large/10;
}
for(passes=0; passes<num; passes++)
{
```

```
for(k=0;k<10;k++)
{
buck[k] = 0;
}
for(i=0;i<n;i++)
{
l = ((arr[i]/div)%10);
bucket[l][buck[l]++] = arr[i];
}
i=0;
for(k=0;k<10;k++)
{
for(j=0;j<buck[k];j++)
{
arr[i++] = bucket[k][j];
}
}
div*=10;
}
}
}
```